Tecnológico De Estudios Superiores De Villa Guerrero

Actividad: Practica 3

EXPLORANDO EL HARDWARE DE TU SISTEMA CON lspci

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Introducción

En esta práctica se utilizó el comando lspci en un sistema Linux para explorar los dispositivos de hardware conectados al sistema mediante el bus PCI (Peripheral Component Interconnect).

El comando permite visualizar información de los dispositivos periféricos y, mediante opciones adicionales, obtener información más detallada sobre los dispositivos y los controladores que utilizan.

Objetivo

Familiarizarse con el comando lspci para identificar y comprender los dispositivos de hardware (periféricos) conectados al sistema a través del bus PCI (Peripheral Component Interconnect)

Desarrollo de la práctica

Para comenzar se abrió la terminal

1. Se ejecutó el siguiente comando:

lspci

El comando lspci permite visualizar los dispositivos de hardware detectados en el sistema mediante el bus PCI.

```
user@debian:~$ lspci
00:00.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne Root Complex
00:00.2 IOMMU: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne IOMMU
00:01.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir PCIe Dummy Host Bridge
00:02.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir PCIe Dummy Host Bridge
00:02.1 PCI bridge: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne PCIe GPP Bridge
00:02.2 PCI bridge: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne PCIe GPP Bridge
00:02.4 PCI bridge: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne PCIe GPP Bridge
00:08.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir PCIe Dummy Host Bridge
00:08.1 PCI bridge: Advanced Micro Devices, Inc. [AMD] Renoir Internal PCIe GPP Bridge to Bus
00:08.2 PCI bridge: Advanced Micro Devices, Inc. [AMD] Renoir Internal PCIe GPP Bridge to Bus
00:14.0 SMBus: Advanced Micro Devices, Inc. [AMD] FCH SMBus Controller (rev 51)
00:14.3 ISA bridge: Advanced Micro Devices, Inc. [AMD] FCH LPC Bridge (rev 51)
00:18.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 0
00:18.1 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 1
00:18.2 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 2
00:18.3 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 3
00:18.4 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 4
00:18.5 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 5
00:18.6 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 6
00:18.7 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 7
01:00.0 SD Host controller: O2 Micro, Inc. SD/MMC Card Reader Controller (rev 01)
02:00.0 Network controller: MEDIATEK Corp. MT7921 802.11ax PCI Express Wireless Network Adapter
03:00.0 Non-Volatile memory controller: Samsung Electronics Co Ltd NVMe SSD Controller PM9C1a (DRAM-less)
04:00.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Lucienne (rev c1)
04:00.1 Audio device: Advanced Micro Devices, Inc. [AMD/ATI] Renoir Radeon High Definition Audio Controller
04:00.2 Encryption controller: Advanced Micro Devices, Inc. [AMD] Family 17h (Models 10h-1fh) Platform Security Processor
04:00.3 USB controller: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne USB 3.1
04:00.4 USB controller: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne USB 3.1
04:00.5 Multimedia controller: Advanced Micro Devices, Inc. [AMD] Audio Coprocessor (rev 01)
04:00.6 Audio device: Advanced Micro Devices, Inc. [AMD] Family 17h/19h/1ah HD Audio Controller
05:00.0 SATA controller: Advanced Micro Devices, Inc. [AMD] FCH SATA Controller [AHCI mode] (rev 81)
05:00.1 SATA controller: Advanced Micro Devices, Inc. [AMD] FCH SATA Controller [AHCI mode] (rev 81)
user@debian:~$
```

Resultado: Después de ejecutar el comando se mostró una lista de los dispositivos de hardware detectados en el sistema. La información permite identificar diferentes componentes y periféricos conectados mediante el bus PCI.

2. Visualización detallada de los dispositivos

Posteriormente se ejecutó el comando:

lspci -v

La opción -v permite obtener información más detallada sobre los dispositivos detectados. Esta información resulta útil para identificar el controlador o módulo del kernel que utiliza cada dispositivo

```
user@debian:~$ lspci -v
00:00.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne Root Complex
        Subsystem: Lenovo Device 3833
        Flags: fast devsel

00:00.2 IOMMU: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne IOMMU
        Subsystem: Lenovo Device 3830
        Flags: bus master, fast devsel, latency 0, IRQ 255
        Capabilities: <access denied>

00:01.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir PCIe Dummy Host Bridge
        Flags: fast devsel, IOMMU group 0

00:02.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir PCIe Dummy Host Bridge
        Flags: fast devsel, IOMMU group 1

00:02.1 PCI bridge: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne PCIe GPP Bridge (prog-if 00 [Normal decode])
        Subsystem: Lenovo Device 394a
        Flags: bus master, fast devsel, latency 0, IRQ 28, IOMMU group 2
        Bus: primary=00, secondary=01, subordinate=01, sec-latency=0
        I/O behind bridge: [disabled] [32-bit]
        Memory behind bridge: fd600000-fd6fffff [size=1M] [32-bit]
        Prefetchable memory behind bridge: [disabled] [64-bit]
        Capabilities: <access denied>
        Kernel driver in use: pcieport

00:02.2 PCI bridge: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne PCIe GPP Bridge (prog-if 00 [Normal decode])
        Subsystem: Lenovo Device 394a
        Flags: bus master, fast devsel, latency 0, IRQ 29, IOMMU group 3
        Bus: primary=00, secondary=02, subordinate=02, sec-latency=0
        I/O behind bridge: [disabled] [32-bit]
        Memory behind bridge: [disabled] [32-bit]
        Prefetchable memory behind bridge: 470200000-4703fffff [size=2M] [32-bit]
        Capabilities: <access denied>
        Kernel driver in use: pcieport

00:02.4 PCI bridge: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne PCIe GPP Bridge (prog-if 00 [Normal decode])
        Subsystem: Lenovo Device 394a
        Flags: bus master, fast devsel, latency 0, IRQ 30, IOMMU group 4
        Bus: primary=00, secondary=03, subordinate=03, sec-latency=0

00:08.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir PCIe Dummy Host Bridge
        Flags: fast devsel, IOMMU group 5

00:08.1 PCI bridge: Advanced Micro Devices, Inc. [AMD] Renoir Internal PCIe GPP Bridge to Bus (prog-if 00 [Normal decode])
        Subsystem: Advanced Micro Devices, Inc. [AMD] Renoir Internal PCIe GPP Bridge to Bus
        Flags: bus master, fast devsel, latency 0, IRQ 31, IOMMU group 6
        Bus: primary=00, secondary=04, subordinate=04, sec-latency=0
        I/O behind bridge: 1000-1fff [size=4K] [16-bit]
        Memory behind bridge: fd100000-fd4fffff [size=4M] [32-bit]
        Prefetchable memory behind bridge: 460000000-4701fffff [size=258M] [32-bit]
        Capabilities: <access denied>
        Kernel driver in use: pcieport

00:08.2 PCI bridge: Advanced Micro Devices, Inc. [AMD] Renoir Internal PCIe GPP Bridge to Bus (prog-if 00 [Normal decode])
        Subsystem: Advanced Micro Devices, Inc. [AMD] Renoir Internal PCIe GPP Bridge to Bus
        Flags: bus master, fast devsel, latency 0, IRQ 32, IOMMU group 7
        Bus: primary=00, secondary=05, subordinate=05, sec-latency=0
        I/O behind bridge: [disabled] [32-bit]
        Memory behind bridge: fd000000-fd0fffff [size=1M] [32-bit]
        Prefetchable memory behind bridge: [disabled] [64-bit]
        Capabilities: <access denied>
        Kernel driver in use: pcieport

00:14.0 SMBus: Advanced Micro Devices, Inc. [AMD] FCH SMBus Controller (rev 51)
        Subsystem: Lenovo Device 386b
        Flags: 66MHz, medium devsel, IOMMU group 8
        Kernel driver in use: piix4_smbus
        Kernel modules: i2c_piix4, sp5100_tco

00:14.3 ISA bridge: Advanced Micro Devices, Inc. [AMD] FCH LPC Bridge (rev 51)
        Subsystem: Lenovo Device 3856
        Flags: bus master, 66MHz, medium devsel, latency 0, IOMMU group 8

00:18.0 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 0
        Flags: fast devsel, IOMMU group 9

00:18.1 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 1
        Flags: fast devsel, IOMMU group 9

00:18.2 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 2
        Flags: fast devsel, IOMMU group 9

00:18.3 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 3
        Flags: fast devsel, IOMMU group 9
        Kernel driver in use: k10temp
        Kernel modules: k10temp

00:18.4 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 4
        Flags: fast devsel, IOMMU group 9

00:18.5 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 5
        Flags: fast devsel, IOMMU group 9

00:18.6 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 6
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        Flags: fast devsel, IOMMU group 9

00:18.7 Host bridge: Advanced Micro Devices, Inc. [AMD] Renoir Device 24: Function 7
        Flags: fast devsel, IOMMU group 9

01:00.0 SD Host controller: O2 Micro, Inc. SD/MMC Card Reader Controller (rev 01) (prog-if 01)
        Subsystem: Lenovo Device 3877
        Flags: bus master, fast devsel, latency 0, IRQ 35, IOMMU group 10
        Memory at fd601000 (32-bit, non-prefetchable) [size=4K]
        Memory at fd600000 (32-bit, non-prefetchable) [size=2K]
        Capabilities: <access denied>
        Kernel driver in use: sdhci-pci
        Kernel modules: sdhci_pci

02:00.0 Network controller: MEDIATEK Corp. MT7921 802.11ax PCI Express Wireless Network Adapter
        Subsystem: Lenovo Device e0bc
        Flags: bus master, fast devsel, latency 0, IRQ 83, IOMMU group 11
        Memory at 470200000 (64-bit, prefetchable) [size=1M]
        Memory at 470300000 (64-bit, prefetchable) [size=16K]
        Memory at 470304000 (64-bit, prefetchable) [size=4K]
        Capabilities: <access denied>
        Kernel driver in use: mt7921e
        Kernel modules: mt7921e

03:00.0 Non-Volatile memory controller: Samsung Electronics Co Ltd NVMe SSD Controller PM9C1a (DRAM-less) (prog-if 02 [NVM Express])
        Subsystem: Samsung Electronics Co Ltd Device a801
        Flags: bus master, fast devsel, latency 0, IRQ 36, IOMMU group 12
        Memory at fd500000 (64-bit, non-prefetchable) [size=16K]
        Capabilities: <access denied>
        Kernel driver in use: nvme
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        Memory at 470304000 (64-bit, prefetchable) [size=4K]
        Capabilities: <access denied>
        Kernel driver in use: mt7921e
        Kernel modules: mt7921e

03:00.0 Non-Volatile memory controller: Samsung Electronics Co Ltd NVMe SSD Controller PM9C1a (DRAM-less) (prog-if 02 [NVM Express])
        Subsystem: Samsung Electronics Co Ltd Device a801
        Flags: bus master, fast devsel, latency 0, IRQ 36, IOMMU group 12
        Memory at fd500000 (64-bit, non-prefetchable) [size=16K]
        Capabilities: <access denied>
        Kernel driver in use: nvme
        Kernel modules: nvme

04:00.0 VGA compatible controller: Advanced Micro Devices, Inc. [AMD/ATI] Lucienne (rev c1) (prog-if 00 [VGA controller])
        Subsystem: Lenovo Device 3f94
        Flags: bus master, fast devsel, latency 0, IRQ 67, IOMMU group 13
        Memory at 460000000 (64-bit, prefetchable) [size=256M]
        Memory at 470000000 (64-bit, prefetchable) [size=2M]
        I/O ports at 1000 [size=256]
        Memory at fd400000 (32-bit, non-prefetchable) [size=512K]
        Capabilities: <access denied>
        Kernel driver in use: amdgpu
        Kernel modules: amdgpu

04:00.1 Audio device: Advanced Micro Devices, Inc. [AMD/ATI] Renoir Radeon High Definition Audio Controller (prog-if 00 [HDA compatible])
        Subsystem: Lenovo Device 3803
        Flags: bus master, fast devsel, latency 0, IRQ 80, IOMMU group 14
        Memory at fd4c8000 (32-bit, non-prefetchable) [size=16K]
        Capabilities: <access denied>
        Kernel driver in use: snd_hda_intel
        Kernel modules: snd_hda_intel

04:00.2 Encryption controller: Advanced Micro Devices, Inc. [AMD] Family 17h (Models 10h-1fh) Platform Security Processor
        Subsystem: Lenovo Device 384c
        Flags: bus master, fast devsel, latency 0, IRQ 34, IOMMU group 15
        Memory at fd300000 (32-bit, non-prefetchable) [size=1M]
        Memory at fd4cc000 (32-bit, non-prefetchable) [size=8K]
        Capabilities: <access denied>
        Kernel driver in use: ccp
        Kernel modules: ccp

04:00.3 USB controller: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne USB 3.1 (prog-if 30 [XHCI])
        Subsystem: Lenovo Device 3860
        Flags: bus master, fast devsel, latency 0, IRQ 58, IOMMU group 16
        Memory at fd100000 (64-bit, non-prefetchable) [size=1M]
        Capabilities: <access denied>
        Kernel driver in use: xhci_hcd
        Kernel modules: xhci_pci

04:00.4 USB controller: Advanced Micro Devices, Inc. [AMD] Renoir/Cezanne USB 3.1 (prog-if 30 [XHCI])
        Subsystem: Lenovo Device 3860
        Flags: bus master, fast devsel, latency 0, IRQ 67, IOMMU group 17
        Memory at fd200000 (64-bit, non-prefetchable) [size=1M]
        Capabilities: <access denied>


      Kernel driver in use: xhci_hcd
      Kernel modules: xhci_pci

04:00.5 Multimedia controller: Advanced Micro Devices, Inc. [AMD] Audio Coprocessor (rev 01)
        Subsystem: Lenovo Device 3855
        Flags: fast devsel, IRQ 79, IOMMU group 18
        Memory at fd480000 (32-bit, non-prefetchable) [size=256K]
        Capabilities: <access denied>
        Kernel modules: snd_pci_acp3x, snd_rn_pci_acp3x, snd_pci_acp5x, snd_pci_acp6x, snd_acp_pci, snd_rpl_pci_acp6x, snd_pci_ps, snd_sof_amd_rembrandt

04:00.6 Audio device: Advanced Micro Devices, Inc. [AMD] Family 17h/19h/1ah HD Audio Controller (prog-if 00 [HDA compatible])
        Subsystem: Lenovo Device 3851
        Flags: bus master, fast devsel, latency 0, IRQ 81, IOMMU group 19
        Memory at fd4c0000 (32-bit, non-prefetchable) [size=32K]
        Capabilities: <access denied>
        Kernel driver in use: snd_hda_intel
        Kernel modules: snd_hda_intel

05:00.0 SATA controller: Advanced Micro Devices, Inc. [AMD] FCH SATA Controller [AHCI mode] (rev 81) (prog-if 01 [AHCI 1.0])
        Subsystem: Advanced Micro Devices, Inc. [AMD] FCH SATA Controller [AHCI mode]
        Flags: bus master, fast devsel, latency 0, IRQ 39, IOMMU group 20
        Memory at fd001000 (32-bit, non-prefetchable) [size=2K]
        Capabilities: <access denied>
        Kernel driver in use: ahci
        Kernel modules: ahci

05:00.1 SATA controller: Advanced Micro Devices, Inc. [AMD] FCH SATA Controller [AHCI mode] (rev 81) (prog-if 01 [AHCI 1.0])
        Subsystem: Advanced Micro Devices, Inc. [AMD] FCH SATA Controller [AHCI mode]
        Flags: bus master, fast devsel, latency 0, IRQ 41, IOMMU group 21
        Memory at fd000000 (32-bit, non-prefetchable) [size=2K]
        Capabilities: <access denied>
        Kernel driver in use: ahci
        Kernel modules: ahci

user@debian:~$
```

Resultado: El comando mostró información detallada de los dispositivos detectados, incluyendo información relacionada con los controladores utilizados por los diferentes componentes del sistema.

5. Identificación del controlador de la tarjeta Wi-Fi

A partir de la información obtenida mediante:

lspci -v

se buscó el dispositivo correspondiente a la tarjeta de red inalámbrica (Wireless).

Tarjeta Wi-Fi: MEDIATEK Corp. MT7921 802.11ax PCI Express Wireless Network Adapter

Driver utilizado por la tarjeta Wi-Fi: mt7921e

6. Identificación del controlador de la tarjeta gráfica

Posteriormente se localizó dentro de la información proporcionada por lspci -v el dispositivo correspondiente a la tarjeta gráfica.

Driver utilizado por la tarjeta grafica: amdgpu

7. Conclusión

Durante esta práctica se utilizó el comando lspci para explorar los dispositivos de hardware presentes en un sistema Linux.

Primero se utilizó lspci para obtener una lista básica de los dispositivos detectados. Posteriormente se utilizó lspci -v para obtener información más detallada y poder identificar los controladores utilizados por dispositivos específicos.

Finalmente, se identificó el controlador correspondiente a la tarjeta de red Wi-Fi y el controlador utilizado por la tarjeta gráfica, cumpliendo con los objetivos establecidos en la práctica.